

Lecture 8

Sampling and Experimental Design

Start with a question

How many hours a week does a typical University of Idaho student spend exercising?

We cannot ask every student. So how would you get the data?

Talk to your neighbor for one minute: who would you ask, and how would you choose them?

Words we need

population

the entire group we want to learn about
every University of Idaho student

parameter

a number that describes the population
the mean weekly exercise hours of ALL UI students

sample

the part of the population we actually observe
the students we end up asking

statistic

a number computed from the sample
the mean exercise hours of the students we asked

More words we need

census

data collected on every member of the population
asking all UI students - slow, expensive, often impossible

sampling frame

the list we actually choose the sample from
the registrar's list of enrolled students

sampling design

the rule we use to choose the sample from the frame

representative sample

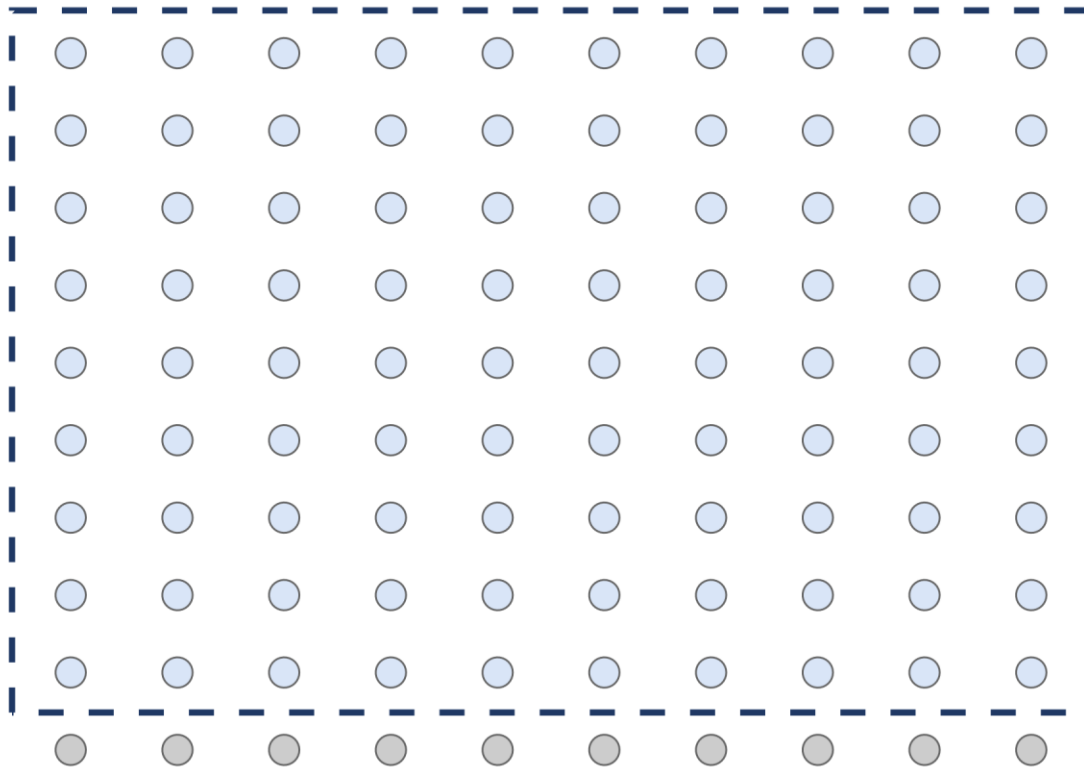
a sample with the same characteristics as the population

sampling error

the gap between a statistic and the parameter, just from which units happened to be chosen
a different sample would give a slightly different answer

Example: exercise at a small college

sampling frame: the registrar's list



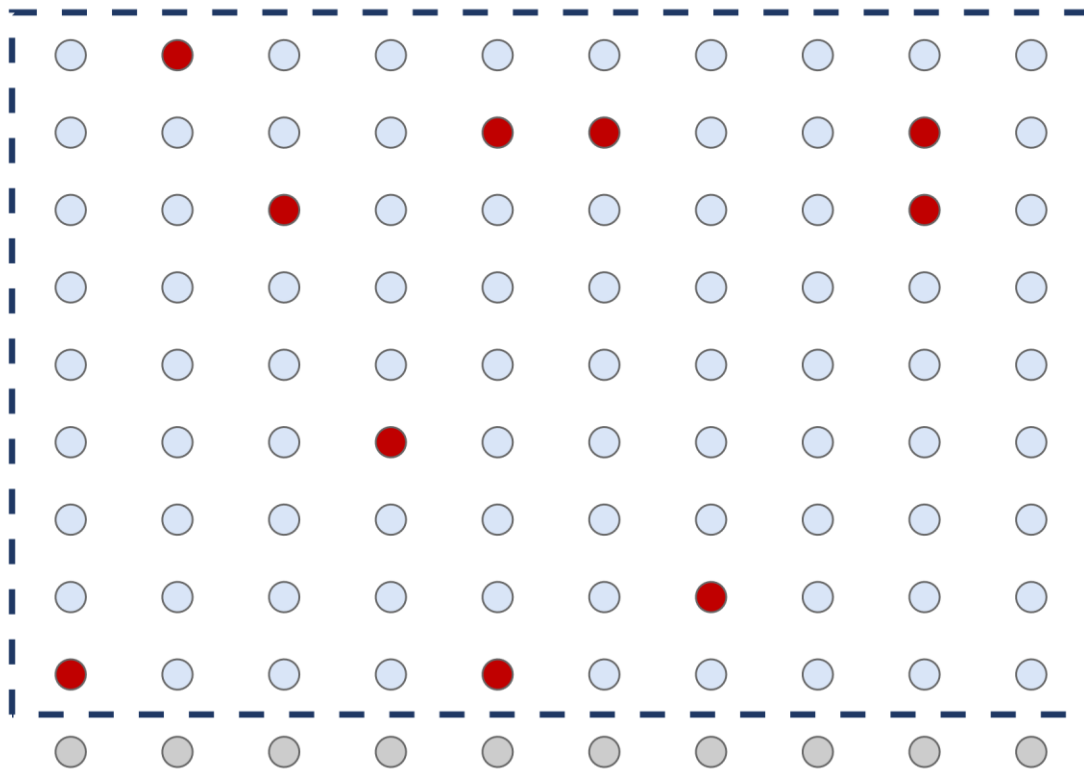
10 late enrollees - not on the list

A small college has 100 students. Its health center wants to know the mean number of hours students exercised last week.

1. What is the population, and what is the parameter?
All 100 students, and the mean exercise hours of all 100.
2. The registrar's list, in the dashed box, has only 90 names. What is that list?
The sampling frame. The 10 late enrollees are not on it, so they cannot be chosen.
3. What would asking all 100 students be called?
A census.

Census, sample and sampling error

sampling frame: the registrar's list



10 late enrollees - not on the list

census

asking all 100 students gives a mean of 4.6 hours - the parameter

sample

an SRS of 10 from the list, in red, gives a mean of 4.4 hours - the statistic

sampling error

$4.4 - 4.6 = -0.2$ hours. A different random sample would give a different error

Practice: find the sampling frame

1. Student government wants to know how students feel about a parking-fee increase. It emails a survey to 500 students picked at random from the university email directory.
Frame: the university email directory. It lists nearly every enrolled student, so it matches the population well.
2. A Moscow coffee shop wants its customers' opinions on a new menu. It surveys 100 people picked at random from its loyalty-app members.
Frame: the loyalty-app member list. Customers who never joined the app cannot be chosen.
3. A city council wants residents' views on a new bike lane. It calls numbers picked at random from a printed phone directory.
Frame: the phone directory. It misses everyone without a listed landline - a group that skews young.

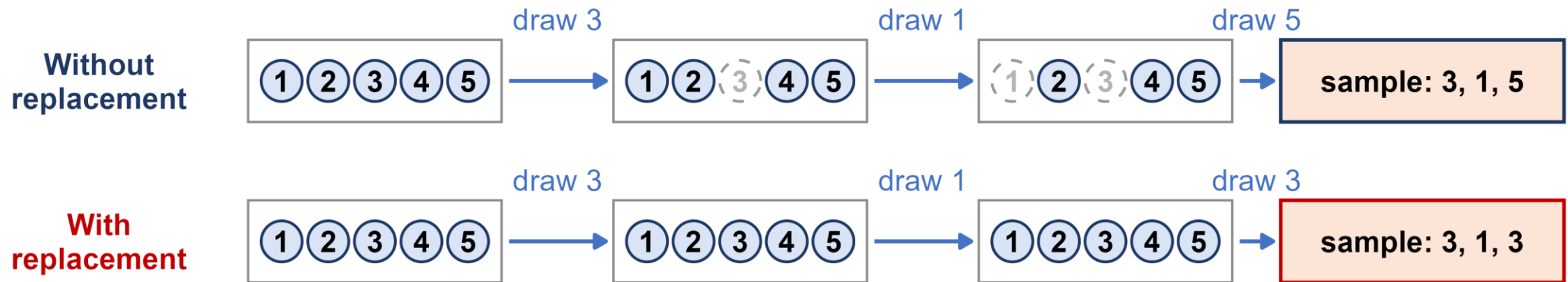
With or without replacement?

sampling without replacement

a picked unit is set aside, so it can be chosen only once

sampling with replacement

a picked unit is put back, so it could be chosen again



Surveys sample without replacement. When the population is much bigger than the sample, the two give practically the same results.

Be Skeptical

Problems with samples

Is the sample representative? A biased sample gives results that are not valid.

Self-selected samples

Only people who chose to respond, as in call-in or online polls - often unreliable.

Sample size

Samples that are too small may be unreliable. Larger is better when possible.

Undue influence

Were the questions asked in a way that pushes people toward an answer?

Nonresponse

Did many people refuse? Those who do answer may have strong opinions.

Causality

A relationship between two variables does not mean one causes the other.

Self-funded studies

Who paid for it, and why? Read it critically - but judge it on its merits.

Misleading use of data

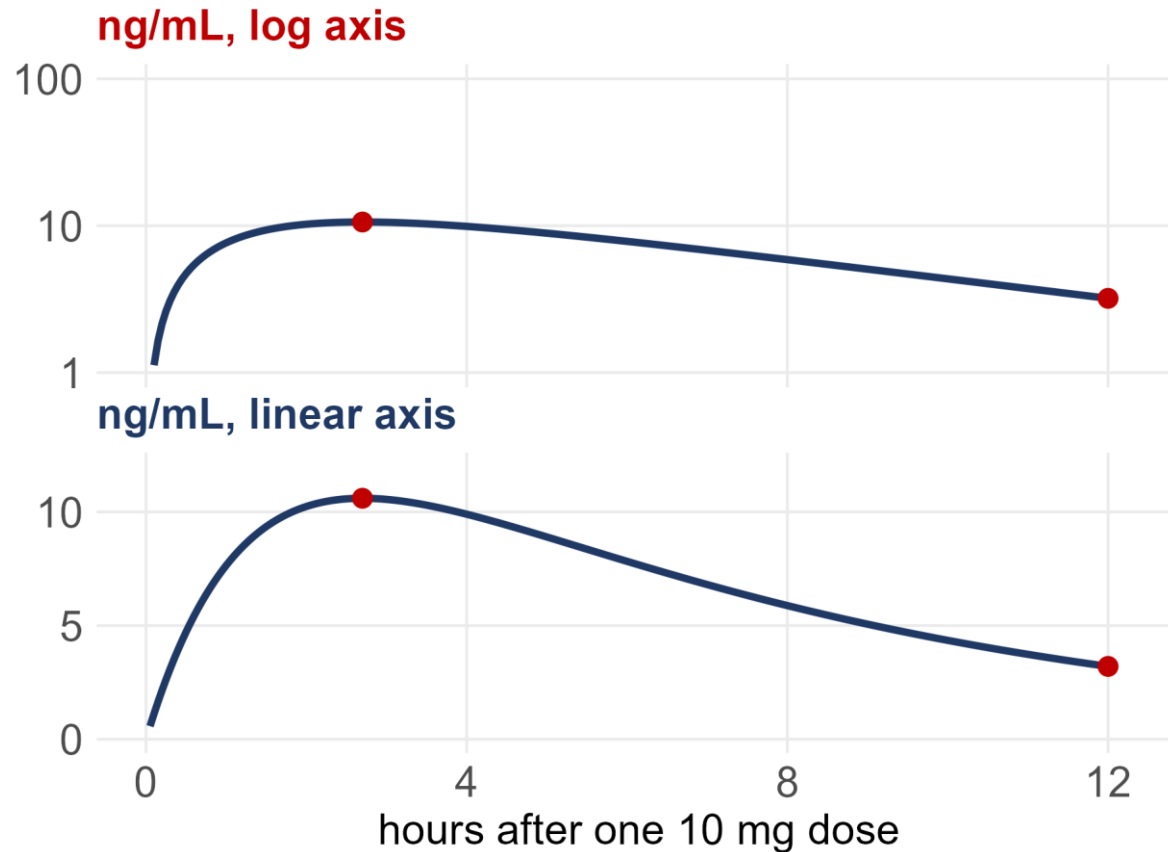
Watch for badly displayed graphs, incomplete data, or missing context.

Confounding

The effects of several factors on the response cannot be separated.

Source: Introductory Statistics 2e (OpenStax), Section 1.2, Critical Evaluation

Misleading graphs: OxyContin



Purdue promoted OxyContin, a 12-hour opioid, with a chart promising “fewer peaks and valleys” in blood levels - drawn on a logarithmic y-axis.

1. Top: log axis. Bottom: the same curve on a linear axis. Which makes the drop look gentle?

The log axis. On the linear axis the level falls from 10.6 to 3.2 ng/mL - about 70% in 12 hours.

In 2007 a Purdue company pleaded guilty to misbranding OxyContin, including sales graphs that exaggerated blood-level differences.

Illustrative curve from FDA label values. Sources: U.S. Attorney, W.D. Va. (2007); CBS News (2011)

Should we believe the graph?

ExxonMobil says its emissions are going down. Its 2024 Advancing Climate Solutions report charts a roadmap for cutting its emissions intensity (emissions per unit produced) - bars that shrink, with no numbers on the y-axis.

1. What can you conclude from bars that shrink but have no numbers?

Only that some cut is planned - not how big it is, or how much rests on “CCS, hydrogen, and/or future advancements.”

2. Exxon's own table: intensity fell from 26.4 to 23.3 tonnes per 100 tonnes produced (2016 to 2023). Did its total emissions fall?

Intensity alone cannot tell you - if output grows, totals can rise. Here the totals did fall: its operations went from 117 to 98 million tonnes.

3. What do those numbers leave out?

Emissions from burning the oil and gas Exxon produces: about 540 million tonnes in 2023, roughly 85% of its footprint, with no target.

Sources: ExxonMobil, 2024 Advancing Climate Solutions; Union of Concerned Scientists (2024)

Two easy ways to get a sample

convenience sample

ask whoever is easiest to reach

stand outside the Student Recreation Center and ask the first 50 people who walk by

voluntary response sample

put out a call and let people choose whether to answer (the textbook calls these self-selected samples)

post a poll on a UI student Instagram account and count the replies

Both are quick, cheap and easy. So what is the problem?

What could go wrong?

For each story: is it a convenience sample or a voluntary response sample? What could go wrong?

1. A nutrition club wants to know how many servings of vegetables UI students eat a day. Members ask the first 60 students who walk out of the campus dining hall at lunch.
Convenience sample. Only lunchtime dining-hall eaters can be chosen - off-campus students and students who cook for themselves are missed.
2. The campus paper wants to know whether students favor moving 8 a.m. classes later. It posts a Vote now! link on social media: 1,200 students vote and 85% say yes.
Voluntary response sample. Students who dread 8 a.m. classes are the most eager to vote, so 85% likely overstates support.
3. A professor wants to know how many hours a week the 300 students in her lecture study. She asks the 25 students who come to her office hours that week.
Convenience sample - just whoever was there. Students who come to office hours probably study more than the class as a whole.

17.5 million votes

On December 18, 2022, Elon Musk posted a poll: “Should I step down as head of Twitter? I will abide by the results of this poll.” In 12 hours it drew more than 17.5 million votes, and 57.5% said yes.

1. Who could vote in this poll?

Anyone who saw the post and chose to vote - mostly Musk's own followers. Nothing stopped fake accounts from voting.

2. What kind of sample is this?

A voluntary response sample.

3. So does 57.5% tell us what Twitter users - or anyone else - think?

No. 17.5 million self-selected votes describe only the people who chose to vote. Size does not fix that.

Sources: NBC News; Associated Press (December 19, 2022)

Online opt-in polls

Many online polls use opt-in panels: anyone can sign up to take surveys, and they are paid for each one they finish.

1. In opt-in polls, what percent of adults under 30 said they were licensed to operate a nuclear submarine?
12% - and 24% of Hispanic respondents. Almost nobody is licensed to operate a nuclear submarine.
2. An opt-in poll found 20% of adults under 30 agreed that the Holocaust is a myth. Pew asked the same question of a random sample. What did it find?
3%. Bogus and careless respondents answering for the money made most of the 20%.
3. Why does this matter for how we read headlines about young people?
Opt-in panels are voluntary response samples. A shocking number about a subgroup may be about who signed up, not what people think.

Source: Mercer, Kennedy and Keeter (2024), Pew Research Center

Estimates are shots at a target

The bullseye is the parameter. Each shot is the estimate from one sample - take a new sample, take a new shot.

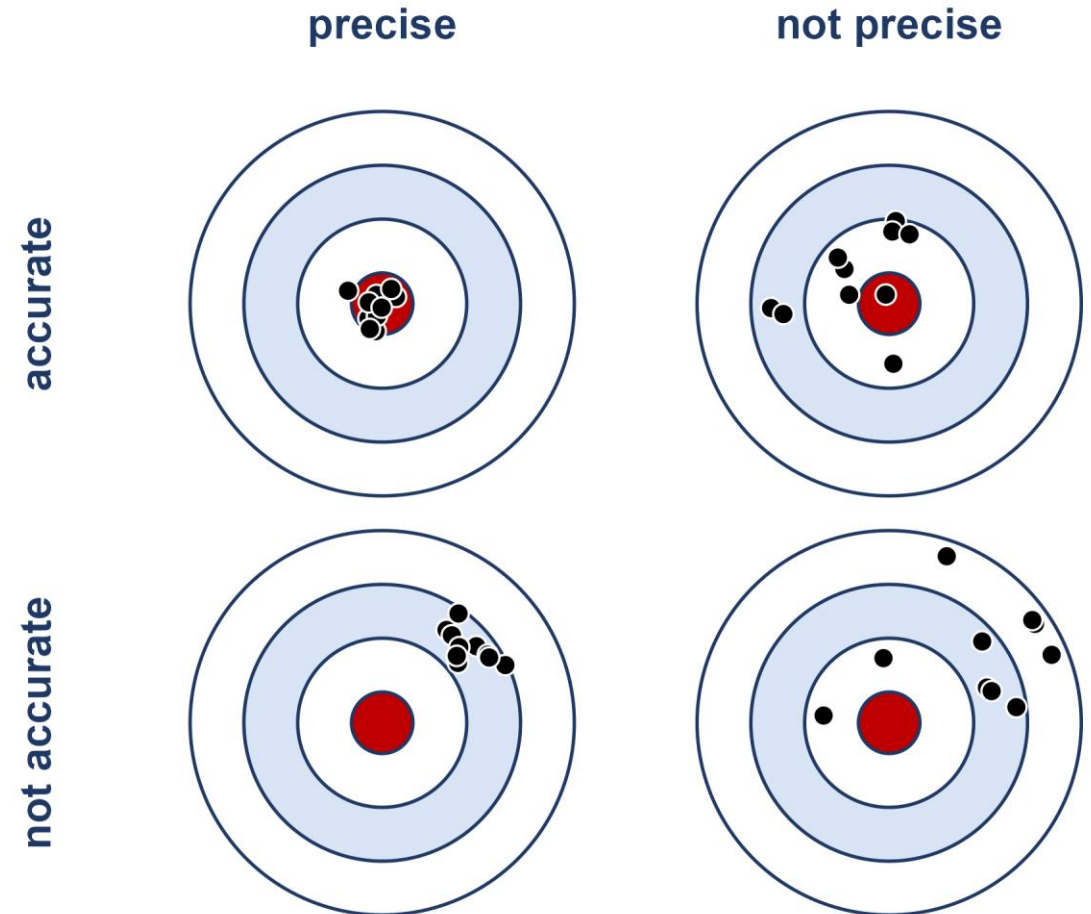
bias

the shots are centered off the bullseye, missing in the same direction every time: an ACCURACY problem

standard error

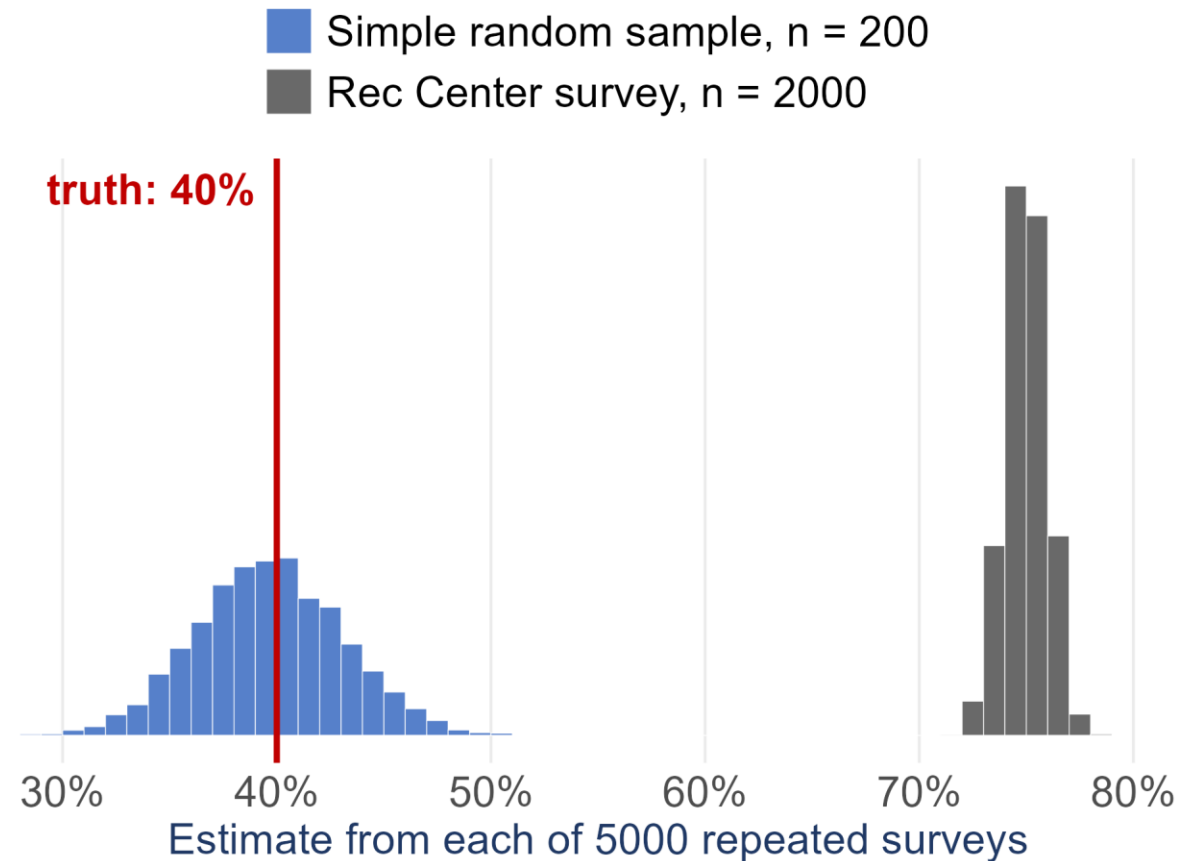
how spread out the shots are from one sample to the next: a PRECISION problem

Convenience and voluntary response samples are biased: they miss the bullseye however carefully we aim.



A bigger sample does not fix bias

- Suppose 40% of UI students exercise at least three times a week
- Blue: 5000 simple random samples of 200 students
- Grey: 5000 Rec Center surveys of 2000 students
- The Rec Center survey is more **precise** - a smaller standard error
- and every one of its estimates misses the truth: **bias**
- More data buys precision. Only a **better design** buys accuracy



The fix: let chance choose

Convenience and voluntary response samples go wrong because a person decides who gets in. Take the person out of it.

random sampling

using chance to choose the sample, so neither the researcher nor the respondents decide who is in it

No person is doing the choosing, so the selection cannot systematically favor anyone - no bias from the selection itself

Chance still moves the estimate from sample to sample, but by rules we understand - so we can measure it. That is the standard error

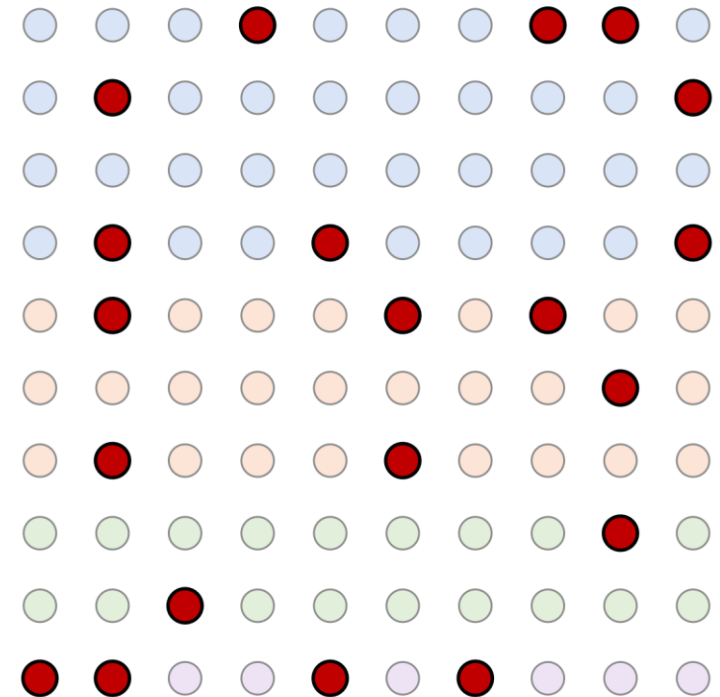
Simple random sample (SRS)

simple random sample (SRS)

every possible group of n units has the same chance of being the sample

Here $N = 100$ students and $n = 20$, so each student has a $20/100 = 1$ in 5 chance of being chosen

This SRS happened to draw 4 seniors but only 2 juniors. An SRS is **fair on average** - not balanced every time



● First-year ● Junior
● Sophomore ● Senior

Drawing an SRS in practice

1. Get a sampling frame and number it 1 to N
2. Let a random number generator choose n different numbers
in R: `sample(1:379, 30)` picks 30 of 379 row numbers
3. The units with those numbers are the sample

`sample()` draws without replacement - no row can be picked twice - unless you add `replace = TRUE`

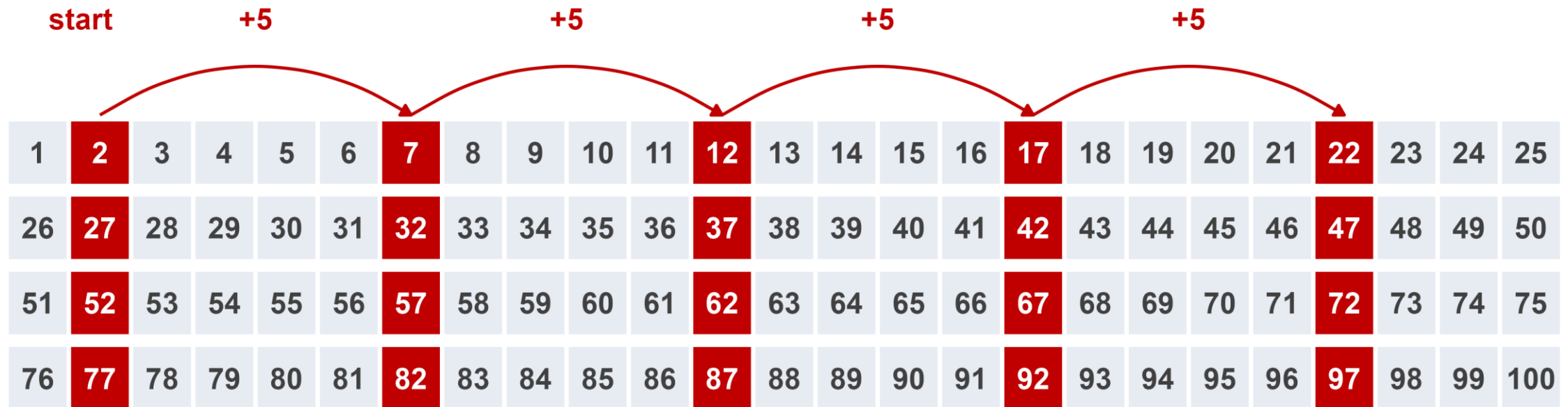
This is exactly what Part 3 of your course project asks you to do.

Systematic sample

systematic sample

pick a random starting point among the first k units of an ordered list, then take every k th unit after it

$k = N/n = 100/20 = 5$. Random start = 2, so we take list positions 2, 7, 12, 17, ... , 97



Works like an SRS - unless the list has a repeating pattern that lines up with k

Stratified random sample

strata

groups of units that are alike, formed BEFORE sampling - class year, region, age group

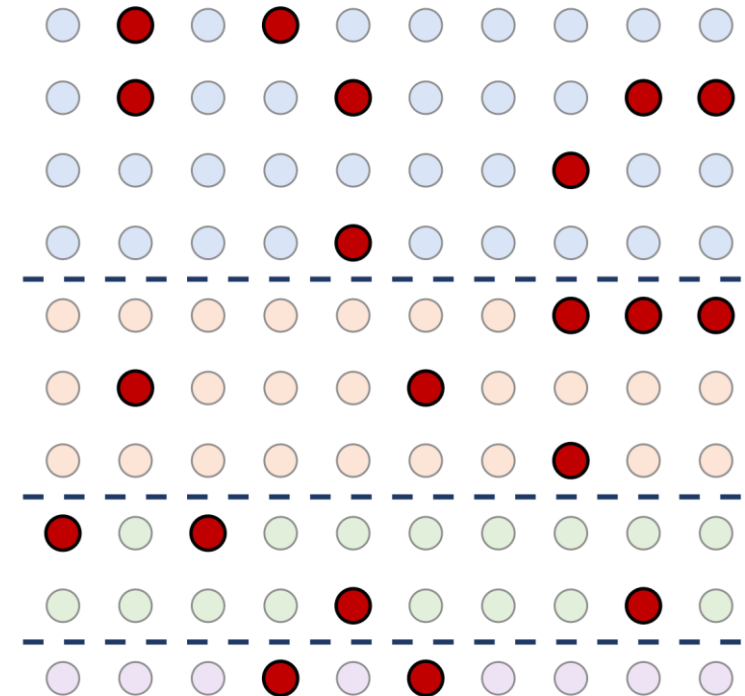
stratified random sample

a separate SRS inside every stratum, combined into one sample

proportional allocation

each stratum gets its share of n : 40, 30, 20, 10 students $\rightarrow$ 8, 6, 4, 2

Every group is guaranteed a place - no more 2 juniors by bad luck



● First-year ● Junior
● Sophomore ● Senior